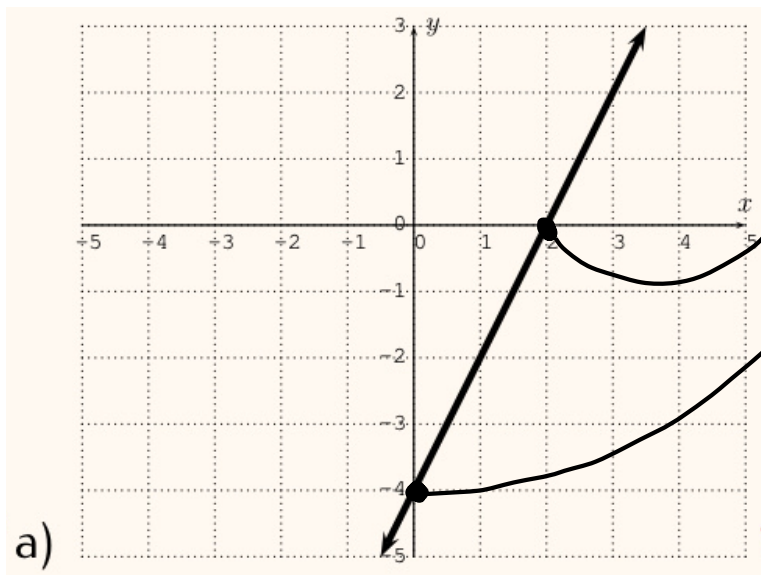


Mat 1375 HW3.

Exercise 3.1

Find the slope and y -intercept of the line with the given data. Using the slope and y -intercept, write the equation of the line in slope-intercept form.

Sol



① Find two points from the graph:

$(2, 0)$ and $(0, -4)$

② Let $(x_1, y_1) = (2, 0)$
 $(x_2, y_2) = (0, -4)$
and find the slope of
line by slope = $\frac{y_2 - y_1}{x_2 - x_1}$

$$\text{slope} = \frac{-4 - 0}{0 - 2} = \frac{-4}{-2} = \frac{4}{2} = 2$$

③ Find the y -intercept point (the intersection point of the graph and y -axis)

y -intercept is $(0, -4)$

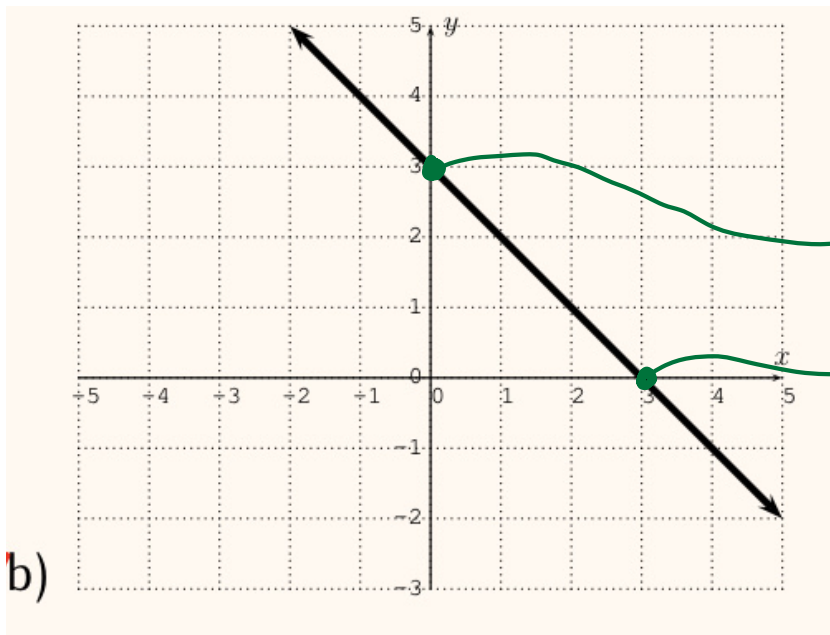
④ Slope-intercept form: $y = (\text{slope}) \cdot x + b$

$y = 2x + b$ and y intercept is $(0, -4)$

plug $(0, -4)$
into $\Rightarrow$

$y = 2x + b$

$$-4 = 2 \cdot 0 + b \Rightarrow b = -4 \Rightarrow y = 2x - 4$$



① Find two points from the graph:

$(0, 3)$ and $(3, 0)$

② Let $(X_1, Y_1) = (0, 3)$
 $(X_2, Y_2) = (3, 0)$

and find the slope

of line by $\frac{Y_2 - Y_1}{X_2 - X_1}$

$$\text{slope} = \frac{0 - 3}{3 - 0} = \frac{-3}{3} = -1$$

③ Find the y-intercept point: $(0, 3)$

④ Find the line equation by slope-intercept form

$$y = (\text{slope}) \cdot x + b$$

By ② and ③, $y = -1 \cdot x + b$ and y-intercept is $(0, 3)$

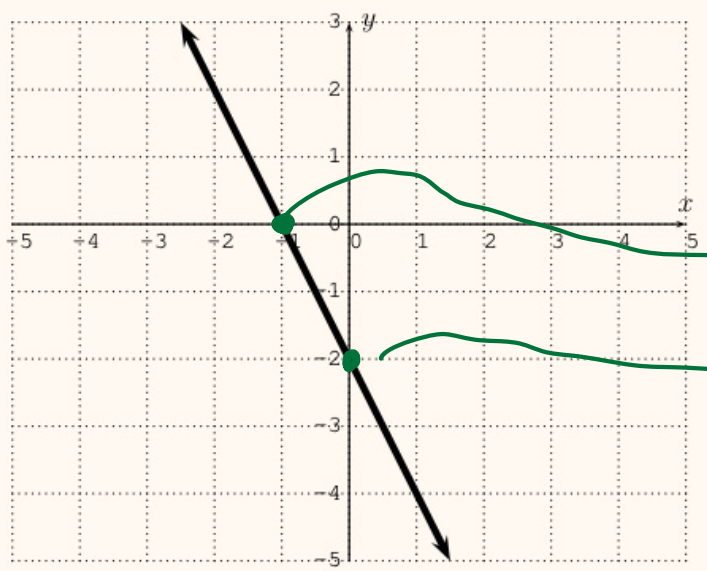
plug $(0, 3)$ into

$$3 = -1 \cdot (0) + b \Rightarrow b = 3$$

$$y = -x + b$$

The line equation is

$$y = -x + 3$$



① Find two points from the graph

$(-1, 0)$ and $(0, -2)$

② Let $(x_1, y_1) = (-1, 0)$
 $(x_2, y_2) = (0, -2)$
 and find the slope of

the line by $\frac{y_2 - y_1}{x_2 - x_1}$

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2 - 0}{0 - (-1)} = \frac{-2}{1} = -2$$

③ Find the y-intercept point: $(0, -2)$

④ Find the line equation by slope-intercept form
 $y = (\text{slope}) \cdot x + b$

By ② and ③, $y = -2x + b$ and y-intercept is $(0, -2)$.

plug $(0, -2)$ into

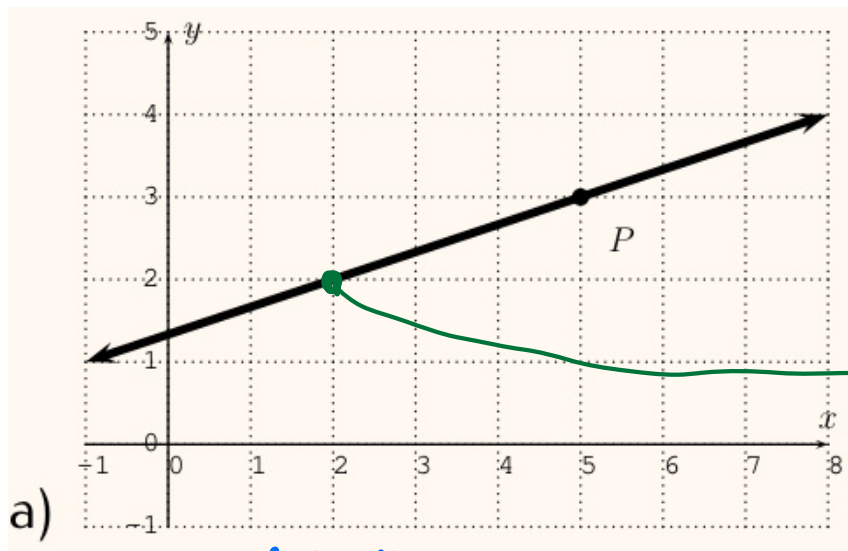
$$\xRightarrow{\substack{\text{plug } (0, -2) \text{ into} \\ y = -2x + b}} -2 = -2 \cdot 0 + b \Rightarrow b = -2$$

The line equation is $y = -2x - 2$

Exercise 3.2

Find the equation of the line in point-slope form (3.3) using the indicated point P .

Sol



① Name the Point P .

P is $(5, 3)$

② Find the other point from graph

$(2, 2)$

③ Let $(x_1, y_1) = (2, 2)$. Find the slope = $\frac{y_2 - y_1}{x_2 - x_1}$

$$\text{slope} = \frac{3 - 2}{5 - 2} = \frac{1}{3}$$

④ Find the equation by point-slope form with $P(5, 3)$

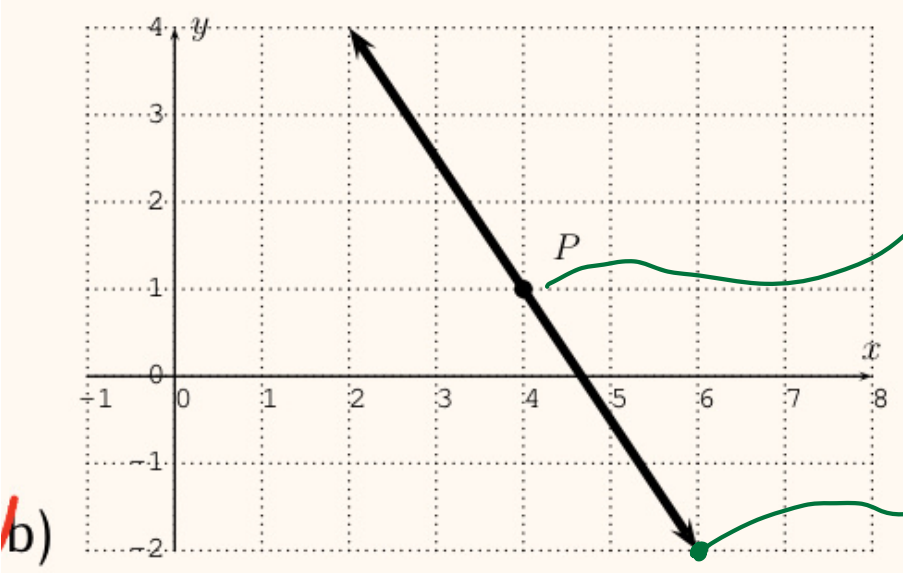
$$\text{slope} = \frac{y - 3}{x - 5}$$

By ③,

The equation is $\frac{1}{3} = \frac{y - 3}{x - 5}$

$$\Rightarrow y - 3 = \frac{1}{3}(x - 5)$$

$$\Rightarrow y = \frac{1}{3}(x - 5) + 3$$



① Name the point P

P is $(4, 1)$

② Find the other point from the graph

$(6, -2)$

③ Let $(x_1, y_1) = (4, 1)$. Find the slope = $\frac{y_2 - y_1}{x_2 - x_1}$
 $(x_2, y_2) = (6, -2)$

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2 - 1}{6 - 4} = \frac{-3}{2}$$

④ Find the line equation by point-slope form with $P(4, 1)$: slope = $\frac{y - 1}{x - 4}$

By ③, the equation is

$$-\frac{3}{2} = \frac{y - 1}{x - 4}$$

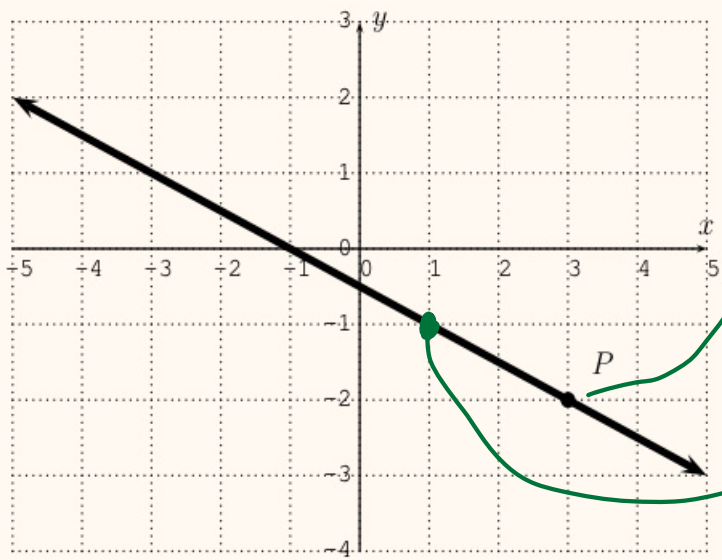
$$\Rightarrow y - 1 = -\frac{3}{2}(x - 4)$$

$$\Rightarrow y = -\frac{3}{2}(x - 4) + 1$$

① Name the point P

P is $(3, -2)$

② Find the point from the graph
 $(1, -1)$



c) ③ Let $(x_1, y_1) = (1, -1)$ and $(x_2, y_2) = (3, -2)$. Find the slope $= \frac{y_2 - y_1}{x_2 - x_1}$

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2 - (-1)}{3 - 1} = \frac{-2 + 1}{2} = -\frac{1}{2}$$

④ Find the line equation by point-slope form with $P(3, -2)$

$$\text{slope} = \frac{y - (-2)}{x - 3}$$

By ③, the line equation is

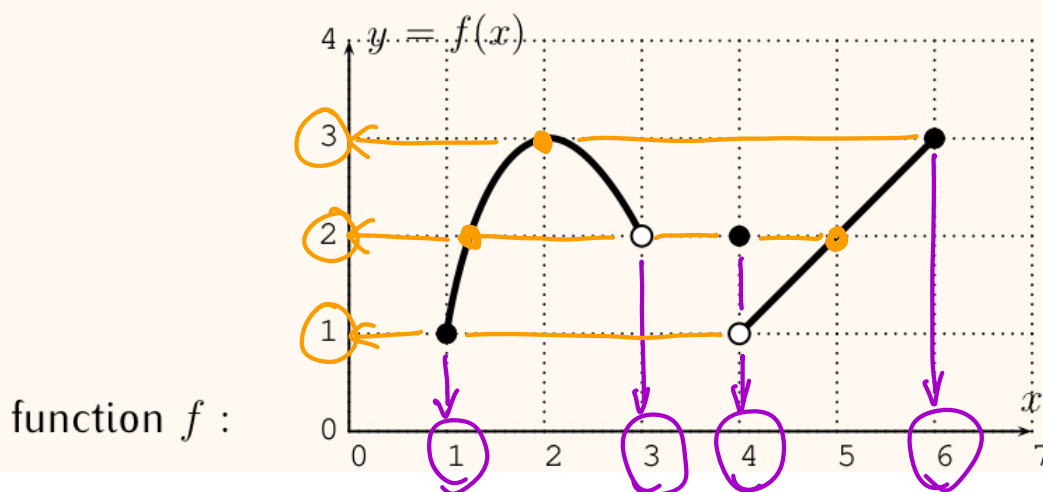
$$-\frac{1}{2} = \frac{y + 2}{x - 3}$$

$$\Rightarrow y + 2 = -\frac{1}{2}(x - 3)$$

$$\Rightarrow y = -\frac{1}{2}(x - 3) - 2$$

Exercise 3.3

Below are three graphs for the functions f , g , and h .



a) Find the domain and range of f .

Sol • For domain of f , we have

① Find the special inputs x that need to discuss

$x=1$ (min. of input, a solid circle)

$x=3$ (an empty circle)

$x=4$ (an empty circle and a solid circle)

$x=6$ (max. of input, a solid circle)

② Discussion of the special points from ①

$x=1 \rightarrow$ solid point

included in domain

$x=3 \rightarrow$ empty point

excluded in domain

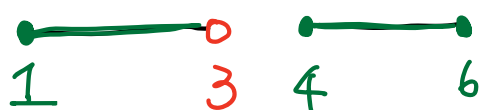
$x=4 \rightarrow$ there is a solid point

included in domain

$x=6 \rightarrow$ solid point

included in domain

③ Domain:



$$\Rightarrow [1, 3) \cup [4, 6]$$

• For range of f , we have

① Find the special outputs y that need to discuss
 $y=1$ (min. of output, a solid circle and an empty circle)
 $y=2$ (3 solid circles and an empty circle)
 $y=3$ (2 solid circles, max of output)

② Discussion of the special points from ①

$y=1 \rightarrow$ it has a solid circle included in range
 $y=2 \rightarrow$ it has 3 solid circles included in range
 $y=3 \rightarrow$ it has 2 solid circles included in range.

③ Range: $\begin{matrix} 3 \\ 2 \\ 1 \end{matrix}$ $\Rightarrow$ Range is $[1, 3]$

✓ d) $f(1)$ ✓ e) $f(2)$ ✓ f) $f(3)$ ✓ g) $f(4)$

Sol

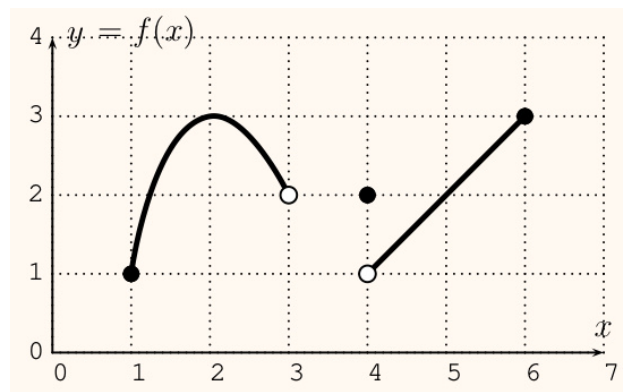
$x=1$ (input) and ask for output

d) $f(1) = 1$

e) $f(2) = 3$

f) $f(3) = \text{undefined}$

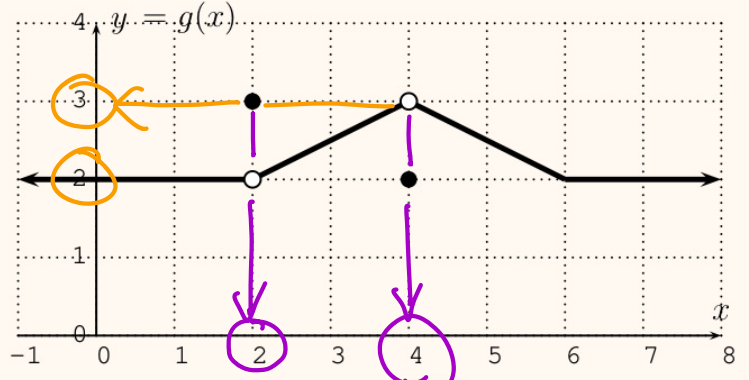
g) $f(4) = 2$



b) Find the domain and range of g .

Sol

- For the domain of g ,
We have



① Find the special inputs x that need to discuss

Arrow to the left: $-\infty$ is the "min."

$x=2$ (an empty circle and a solid circle)

$x=4$ (a solid circle and an empty circle)

Arrow to the right: ∞ is the "max."

② Discussion:

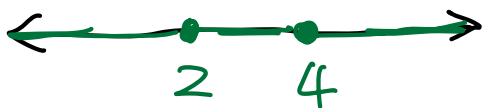
$x=2 \rightarrow$ it has a solid circle

included in domain

$x=4 \rightarrow$ it has a solid circle

included in domain

③ Domain



$\Rightarrow$

domain is $(-\infty, \infty)$

- For the range of g , we have

① Find the special outputs y that need to discuss.

$y=2$ (infinitely many solid circle and an empty circle)

$y=3$ (one solid circle and an empty circle)

② Discussion

$y=2 \rightarrow$ it has infinitely many solid circle

included.

$y=3 \rightarrow$ it has one solid circle

included.

③ Range $\begin{matrix} 3 \\ \downarrow \\ 2 \end{matrix} \Rightarrow \boxed{\text{range is } [2, 3]}$

✓ m) $g(2)$ ✓ n) $g(3)$ ✓ o) $g(4)$ ✓ p) $g(6)$ ✓ q) $g(13.2)$

Sol. $x=2$ (input) and ask for output

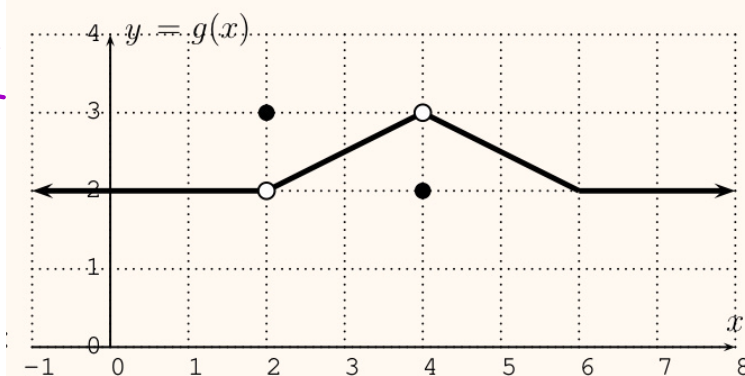
$$g(2) = 3$$

$$g(3) = 2.5$$

$$g(4) = 2$$

$$g(6) = 2$$

$$g(13.2) = 2$$



✓ c) Find the domain and range of h .

Sol. • For the domain of h , we have

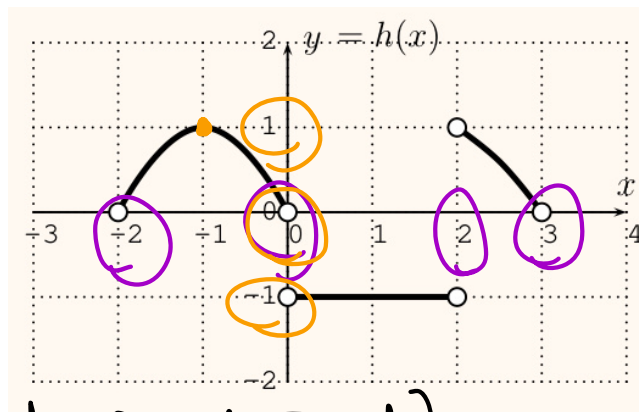
① Find the special inputs x that need to discuss:

$x = -2$ (one empty circle and min. of input)

$x = 0$ (two empty circles)

$x = 2$ (two empty circles)

$x = 3$ (one empty circle and max. of input)



② Discussion:

$x = -2 \rightarrow$ it has **no** solid circle

excluded

$x = 0 \rightarrow$ it has **no** solid circle

excluded

$x = 2 \rightarrow$ it has **no** solid circle

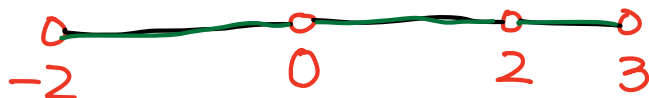
excluded

$x = 3 \rightarrow$ it has **no** solid circle

excluded

③ Domain

$$\Rightarrow (-2, 0) \cup (0, 2) \cup (2, 3)$$



- For the range of h , we have

① Find the special outputs y that need to discuss

$y = 1$ (max of output and one empty circle, one solid circle)

$y = 0$ (3 solid circles)

$y = -1$ (min of output and infinitely many solid points and 2 empty circles)

② Discussion

$y = 1 \rightarrow$ it has a solid circle

Included

$y = 0 \rightarrow$ it has **no** solid circle

excluded

$y = -1 \rightarrow$ it has solid circles

included



$$\Rightarrow R_{h(x)} = \{x \mid x \in \{-1\} \text{ or } x \in (0, 1]\}$$

• -1

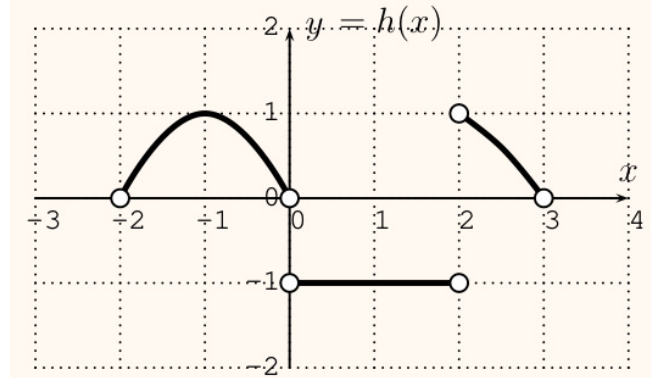
✓ r) $h(-2)$ ✓ s) $h(-1)$ ✓ t) $h(0)$

Sol

$h(-2) = \text{undefined}$

$h(-1) = 1$

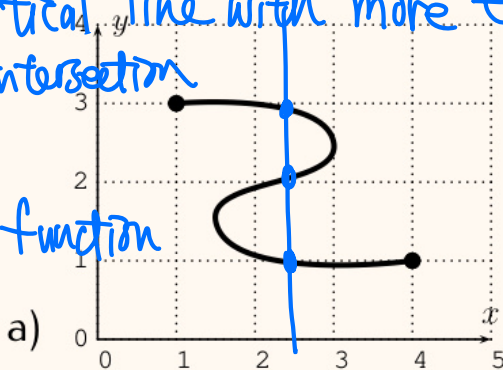
$h(0) = \text{undefined}$



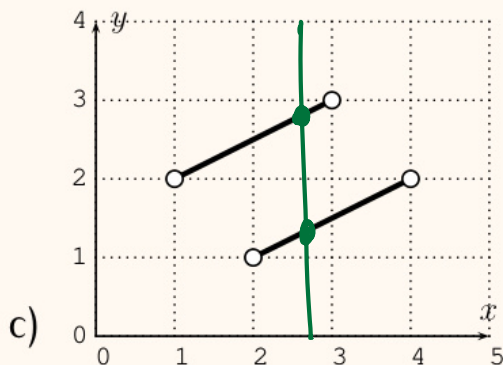
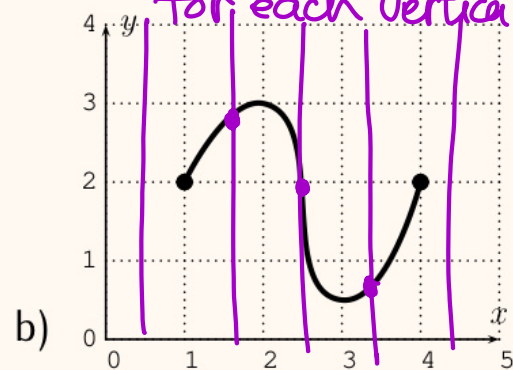
✓ Exercise 3.4

Use the vertical line test to determine which of the following graphs are the graphs of functions.

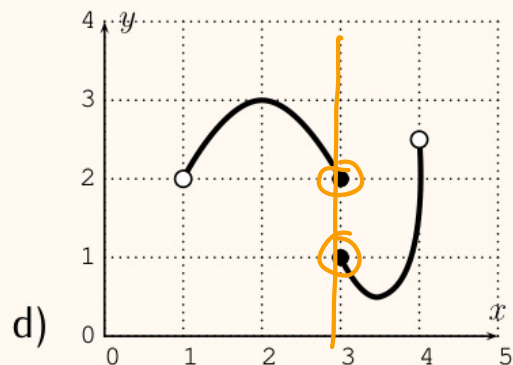
a) a vertical line with more than one intersection point
 $\Rightarrow$ NOT a function



b) At most one intersection point for each vertical line
 $\Rightarrow$ it's a function



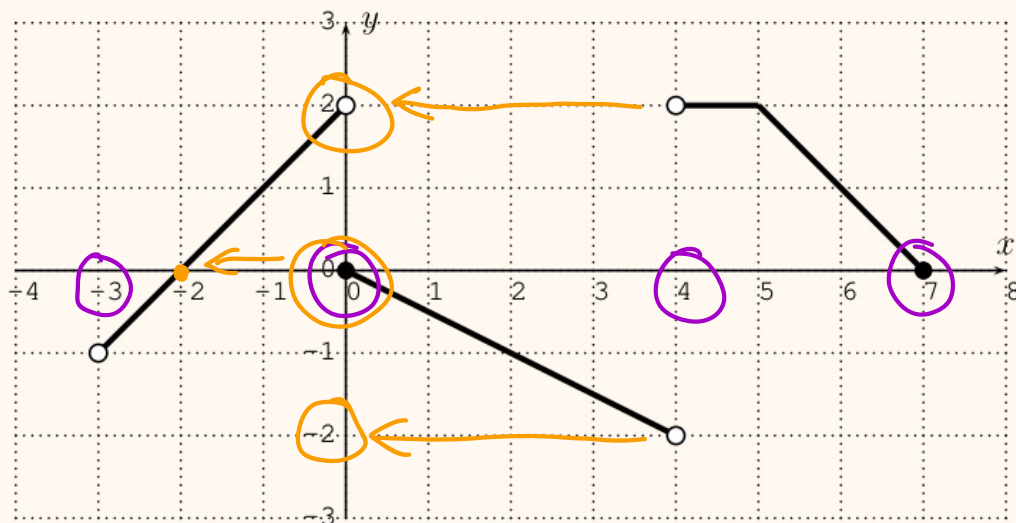
c) a vertical line with more than one intersection point
 $\Rightarrow$ Not a function



d) a vertical line with more than one intersection point
 $\Rightarrow$ NOT a function

Exercise 3.5

Let f be the function given by the following graph.



- a) What is the domain of f ? b) What is the range of f ?

Sol

a) For the domain of f , we have

① Find the special inputs x that need to discuss

$x = -3$ (min. of input and an empty circle)

$x = 0$ (an empty circle and a solid circle)

$x = 4$ (two empty circles)

$x = 7$ (max of input and a solid circle)

② Discussion

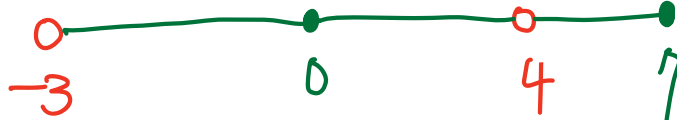
$x = -3 \rightarrow$ it has no solid circle $\Rightarrow$ excluded

$x = 0 \rightarrow$ it has a solid circle $\Rightarrow$ included

$x = 4 \rightarrow$ it has no solid circle $\Rightarrow$ excluded

$x = 7 \rightarrow$ it has a solid circle $\Rightarrow$ included

③ Domain



$\Rightarrow$ domain is

$$(-3, 4) \cup (4, 7)$$

b) For the range of f , we have

① Find the special outputs y that need to discuss.

$y=2$ (two empty circle and infinitely many solid circles and max of output)
 $y=0$ (three solid point)
 $y=-2$ (min of output and an empty circle)

② Discussion:

$y=2 \rightarrow$ it has solid circles $\rightarrow$ included

$y=0 \rightarrow$ it has solid circles $\rightarrow$ included

$y=-2 \rightarrow$ it has no solid circle. $\rightarrow$ excluded

③ Range: $\begin{matrix} 2 \\ 0 \\ -2 \end{matrix}$

$\Rightarrow$ the range is $[-2, 2]$.

c) For which x is $f(x) = 0$?

d) For which x is $f(x) = 2$?

e) For which x is $f(x) \leq 1$?

f) For which x is $f(x) > 0$?

g) Find $f(2)$ and $f(5)$.

h) Find $f(2) + f(5)$.

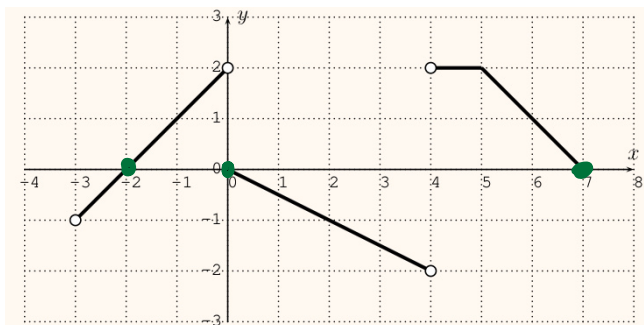
i) Find $f(2) + 5$.

j) Find $f(2 + 5)$.

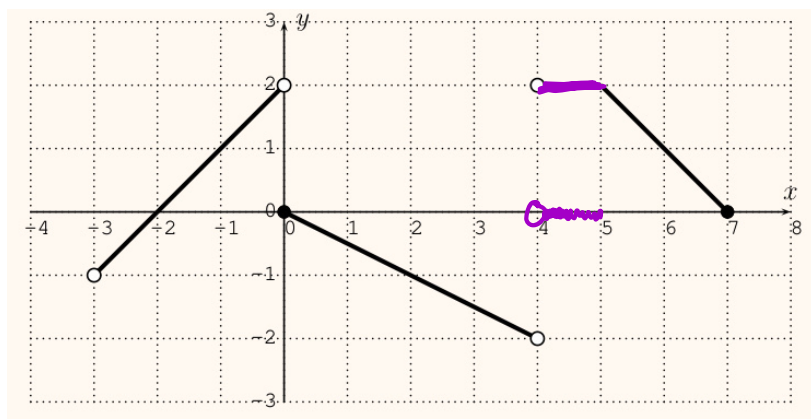
Sol

(c) Find the input(s) x such that the output(s) y is 0 ($f(x)=0$).

$x = -2, x = 0, x = 7$



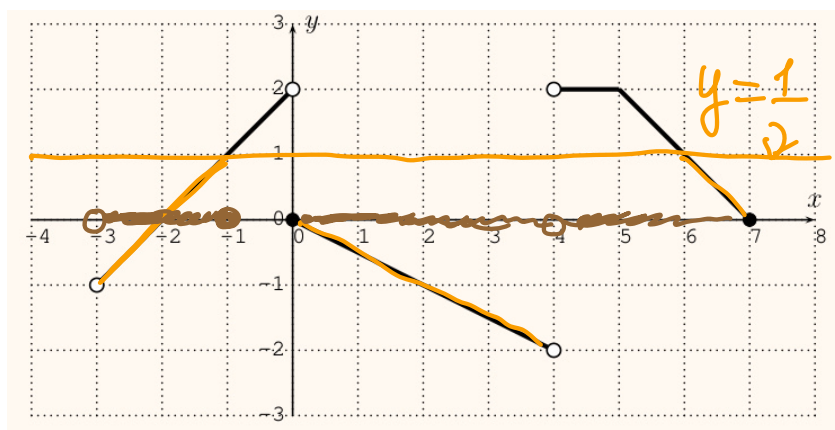
(d) Find the inputs x such that the outputs y is 2
 $(f(x)=2)$



$$x \in (4, 5]$$

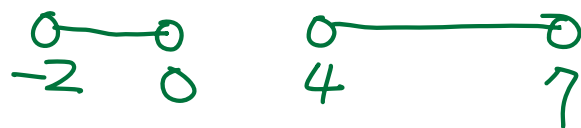
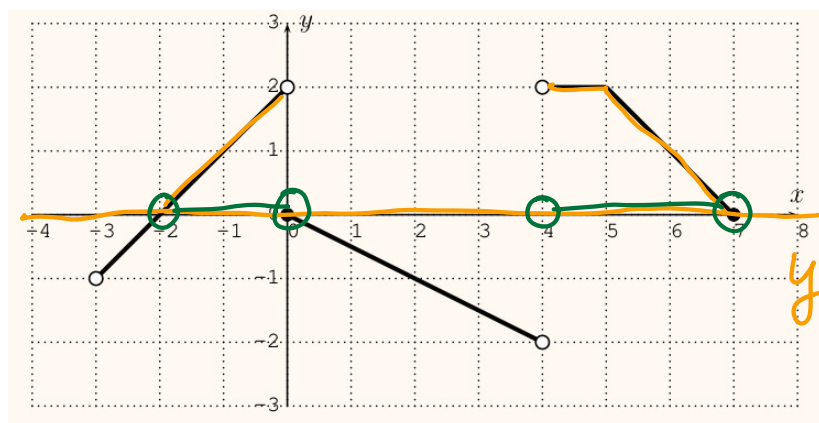
(it is not just a point but an interval)

(e) Find the inputs x such that the outputs $y \leq 1$
 $(f(x) \leq 1)$



$$x \in [-3, -1] \cup [0, 4]$$

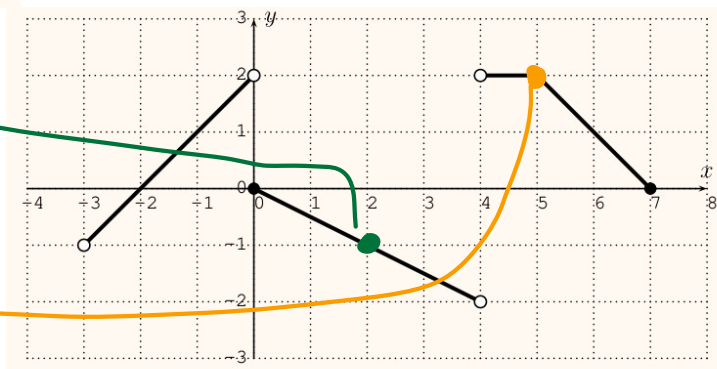
~~(f)~~ Find the inputs x such that the outputs $y > 0$
 $(f(x) > 0)$



$$x \in (-2, 0) \cup (4, 7)$$

(g) $f(2) = -1$ ←

$f(5) = 2$ ←



$$(b) f(2) + f(5) = -1 + 2 = 1$$

$$(i) f(2) + 5 = -1 + 5 = 4$$

$$(j) f(2+5) = f(7) = 0$$

